YATHARTH GOYAL

SOFTWARE DEVELOPMENT / JAVA / FULL-STACK ENGINEERING / APPLIED AI

Jaipur, India · yatharthgoyal657@gmail.com · **GitHub** · **LinkedIn** · **LeetCode**

TECHNICAL SKILLS

Languages Java, JavaScript, Python, SQL, HTML, CSS

Backend & Frameworks Express.js, Node.js, FastAPI, REST APIs, React, Vite, Tailwind CSS

Data & Infrastructure PostgreSQL, PostGIS, Redis / BullMQ, Docker, GitHub Actions, Render, AWS SAM (templates)

Testing & AI Git, pytest, Playwright, scikit-learn, YOLO, NumPy, pandas, SciPy / HiGHS, MILP

SELECTED ENGINEERING PROJECTS

AegisED | Emergency Operations & Decision Support

REPO · **LIVE DEMO**

React · Express · PostgreSQL · FastAPI · scikit-learn · Render / Neon

- Engineered database-backed emergency intake, doctor/bed allocation and discharge workflows with PostgreSQL transactions and row-level locking to protect concurrent resource assignments.
- Implemented role-scoped authorization, server-side session protections and transactional audit logging; integrated a human-reviewed FastAPI ML service with recorded overrides.
- Compared four ML candidates; documented macro-F1 0.7893 on 900 held-out synthetic cases (not clinically validated). The repository includes API, ML and browser-testing evidence.

EvalForge | Distributed Assessment Intelligence · PRIVATE · RELEASE PENDING

React · Express · PostgreSQL · Redis / BullMQ · FastAPI · Docker

- Built versioned rubric and submission workflows with criterion-level evidence, instructor approval/overrides, student appeals and organization-scoped access controls.
- Separated API, job dispatcher, queue workers and evaluator; added durable PostgreSQL jobs with idempotency, bounded retries and dead-letter recovery. Local checks: 15/15 backend and 7/7 evaluator tests.

Grid Flex | Industrial Energy Scheduling Optimizer

SOURCE

Python · SciPy / HiGHS · Mixed-Integer Optimization · JavaScript · AWS SAM

- Modeled machine scheduling, solar use, battery dispatch and grid limits in a MILP with distinct cost, balanced and carbon-priority strategies across synthetic operating scenarios.
- Created an independent constraint verifier, stress tests and battery-sensitivity tools; documented 52/52 local tests passing. AWS deployment is prepared but not live-verified.

RouteGuard 2.0 | Road-Hazard Intelligence · IN PROGRESS

Python / YOLO · React / Leaflet · Express · PostgreSQL / PostGIS · TomTom

- Developing road-damage detection evaluation and hazard-informed navigation; auditing labels, dataset splits and model errors while integrating geospatial route comparison.
- Research goal: ≥95% precision AND recall per severe-hazard class on independently verified data. Goal not yet achieved; experimental models are not production validated.

JAVA PROBLEM SOLVING 16-topic DSA repository with 161 catalogued LeetCode implementations (repository coverage, not verified accepted submissions). **VIEW CODE**

EDUCATION

B.Tech in Information Technology | JECRC, Jaipur — **Expected 2028**